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Rogex PRISMA 3

Technical Whitepaper v1.2

Subject-normalized EEG decoding software for research

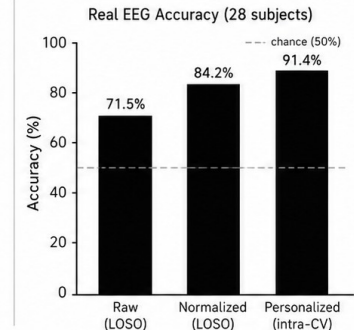
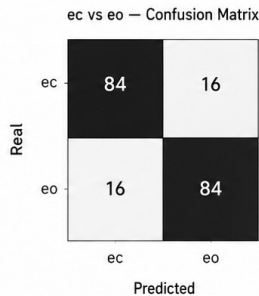


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PRISMA 3

Real EEG validation

OpenNeuro DS007358 • 28 subjects • ec vs eo



84.2%
accuracy
normalized (LOSO)



+12.7 pts
vs raw features



91.4%
personalized
(intra-CV)



SUBJECT-NORMALIZED EEG DECODING

PUBLIC TECHNICAL RELEASE

Public technical release graphic: PRISMA 3 real EEG validation summary.

Field	Value
Author and initial rights holder	Roger Navarro
Research line	Rogex Laboratories / Rogex PRISMA
Release	PRISMA 3 Research Core - Public Technical Release
Status	Experimental research software; non-diagnostic; non-medical
Primary claim	Subject-normalized EEG decoding improves method-level EEG interpretation
License	Rogex PRISMA Research Source License v1.0: free research use; no commercial resale/embedding without permission

Important scientific boundary

PRISMA 3 is not a medical device and does not diagnose, treat, cure, or predict disease. Results are research-method evidence, not clinical performance claims.

1. Executive summary

Rogex PRISMA 3 is an experimental EEG research software framework created by Roger Navarro under the Rogex Laboratories research line. It is designed for signal ingestion, preprocessing, feature extraction, personal baseline modelling, subject-relative translation, uncertainty estimation, machine-learning evaluation, and reproducible reporting.

The central technical idea is simple: EEG should not be interpreted only through universal thresholds. Every subject has a different neurophysiological signature. PRISMA 3 therefore interprets signal features relative to each user baseline and uncertainty profile.

This public whitepaper consolidates the technical architecture, real EEG validation status, synthetic demonstration results, scientific limits, and licensing terms for public release. It is written to be useful for researchers, collaborators, reviewers, and potential institutional partners.

Core thesis

Model the person before interpreting the state. The individual baseline is treated as part of the signal, not as noise to ignore.

2. Real EEG validation: OpenNeuro DS007358

PRISMA 3 includes a supervised validation appendix over real EEG data using OpenNeuro DS007358, a resting-state dataset from India and Tanzania. The evaluated task is eyes closed versus eyes open (ec vs eo), a robust EEG contrast associated with occipital alpha blocking.

In this validation run, 28 subjects were used after retaining montages with occipital channels and discarding frontal-only montages below the channel threshold. The evaluation uses 3,304 EEG windows and Leave-One-Subject-Out (LOSO) validation across subjects. Chance level for the task is 50%.

Configuration	Accuracy	F1	Interpretation
Global raw features (LOSO)	71.5%	68.4	Between-subject baseline using raw features
Global subject-normalized (LOSO)	84.2%	84.1	Subject-relative normalization improves generalization
Personalized per-user model (intra-CV)	91.4%	-	Easier within-subject regime; not equivalent to cross-subject generalization

Key measured gain

Subject normalization adds +12.7 percentage points over raw features in the real EEG ec/eo validation. This supports PRISMA 3's central method claim: inter-individual variability should be modelled rather than ignored.

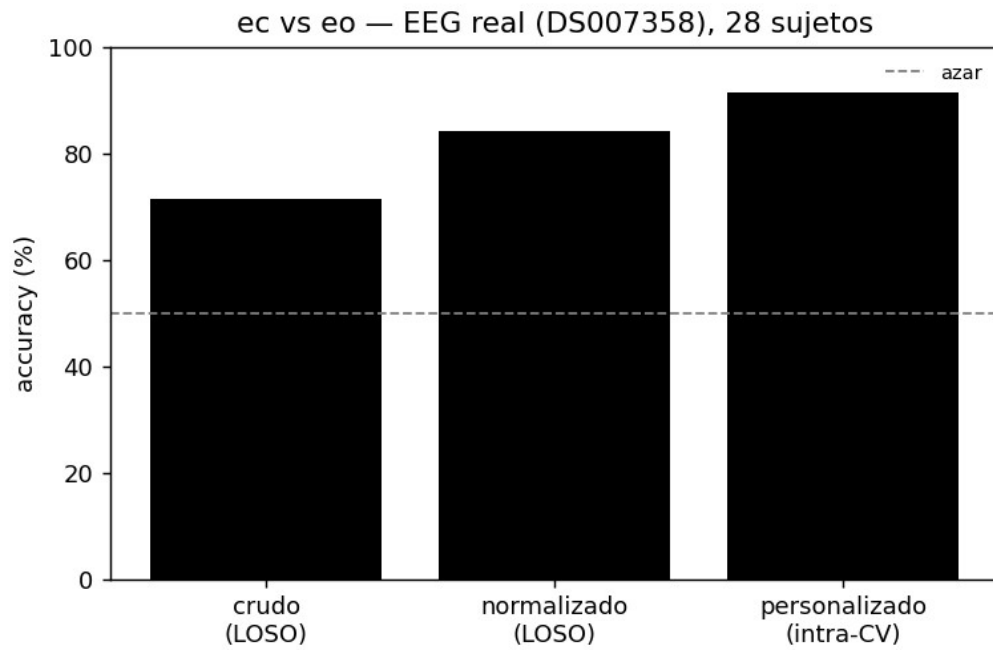


Figure 1 - Accuracy comparison: raw LOSO, subject-normalized LOSO, and personalized intra-CV.

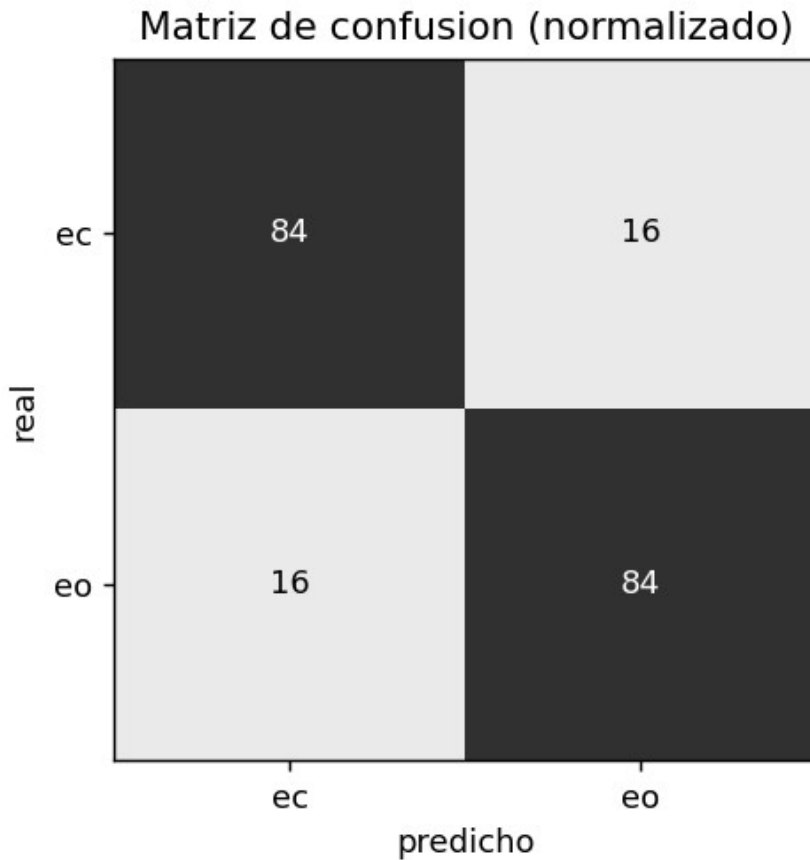


Figure 2 - Normalized confusion matrix for ec vs eo classification.

Physiological check: the validation report observes occipital relative alpha of $ec=0.338$ versus $eo=0.083$, matching the expected increase in alpha during eyes-closed resting state. This matters because the classifier is aligned with a known EEG effect rather than only optimizing a numerical score.

3. Evidence layers and responsible interpretation

Layer	Status	Interpretation
Synthetic supervised demo	Implemented	Five-state synthetic-user benchmark validates pipeline mechanics and personalization strategy; not clinical EEG performance.
Real EEG ingestion	Implemented	CSV, MNE-compatible files and eegdash/OpenNeuro pathways can be transformed into PRISMA features.
Real EEG supervised ec/eo appendix	Implemented	DS007358 ec/eo task evaluated with LOSO across 28 subjects; validates method on a robust EEG contrast.
Clinical claims	Not claimed	No diagnosis, disease prediction, treatment, or medical decision support.

The real ec/eo task is intentionally chosen as a first validation target because alpha blocking is one of the most reliable EEG phenomena. This does not automatically generalize to difficult affective, cognitive, clinical, or spiritual-state classification tasks. It shows that the method and software pipeline can work on real EEG under a physiologically grounded condition.

4. Technical architecture

PRISMA 3 follows the pipeline below:

EEG input -> preprocessing -> windows -> features -> personal baseline -> individual translator -> uncertainty/confidence -> ML evaluation -> explainable report

Module	Function
src/core/eeg_simulator.py	Synthetic EEG generator with user variability, IAF-centered alpha and artifacts.
src/core/feature_extraction.py	Bandpower, ratios, spectral entropy, Hjorth parameters, RMS, variance, artifact score and signal quality metrics.
src/core/baseline_engine.py	Personal baseline computation, z-scores and within-subject deviation.
src/core/individual_translator.py	Subject-relative interpretation, adaptive thresholds, confidence, uncertainty and drift logic.
src/ml/	Dataset construction, training, LOSO evaluation, prediction and personalization.
src/utis/io.py	Real EEG ingestion using MNE-compatible raw data and file formats.
analyze_bids.py / analyze_eo_ec.py	Real-dataset validation and BIDS-style task evaluation pathways.
run_app.py	Streamlit interface for real EEG, simulation, baseline, results and ML pages.

5. Method: subject-normalized EEG decoding

Raw EEG features combine at least two components: the state-related effect of interest and the subject-specific physiology/montage/profile. Universal thresholds often confuse these components.

PRISMA 3 builds personal baselines and expresses new readings as deviations relative to the same subject. This makes state changes more comparable across people while preserving the fact that each user has a unique signature.

The result is not a claim that personal models are always sufficient. Instead, it is a practical engineering strategy: use global models as starting points, but prioritize individual calibration where enough user data exists.

6. Conceptual software mapping

Symbol	Meaning	Software representation
Is	Surviving information	Extracted EEG features that remain available after filtering and noise/artifact handling.
A	Absence / uncertainty	Signal loss, missing data, artifacts, low signal quality and model uncertainty.
E	State / context	Task, exercise, report, body state or experimental condition.
Mp	Personal memory	Baseline, prior sessions, individual variability and response history.
Tb	Individual translator	Engine that interprets features relative to baseline and uncertainty.
Sp	Estimated state	Probable state, confidence, uncertainty and explanation.

7. Synthetic demonstration benchmark

The package also includes a synthetic supervised benchmark over five neurophysiological state labels. These values validate the separability of the simulator and the pipeline mechanics; they are not clinical or real-world EEG performance numbers.

Configuration	Accuracy	Interpretation
Global raw features (LOSO)	76.7%	Between-subject baseline on synthetic demo data
Global normalized by user (LOSO)	94.4%	Shows normalization benefit in the synthetic benchmark
Personalized intra-user CV	96.1%	Shows personalized-model priority in the synthetic benchmark

8. Reproducibility and public repository guidance

The public repository should include README.md, LICENSE, docs/, outputs/reports/, figures, and clear commands for running the demo, training/evaluation, and real EEG analysis scripts.

Recommended public wording: 'PRISMA 3 is research software for subject-normalized EEG analysis. It supports simulated and imported real EEG data. It is non-diagnostic and not a medical device.'

The repository should avoid including large datasets, private recordings, cached OpenNeuro directories, macOS metadata, Python bytecode, model artifacts not intended for release, or personally identifying data.

Task	Command
Install environment	<code>python -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt</code>

Run Streamlit UI	<code>streamlit run run_app.py</code>
Run synthetic demo	<code>python demo_prisma3.py</code>
Train/evaluate demo model	<code>python train_model.py && python evaluate_model.py</code>
Analyze EEGDash/OpenNeuro-style data	<code>python analyze_eegdash.py --dataset NM000147 --cache ./data --n 3</code>
Analyze local BIDS dataset	<code>python analyze_bids.py --path ./data/dataset --n 3</code>

9. Licensing and intellectual property position

PRISMA 3 is released under the Rogex PRISMA Research Source License v1.0. This is a source-available research license, not a permissive open-source license. Research, personal, and academic non-commercial use are allowed. Commercial resale, commercial embedding, SaaS monetization, or selling products containing PRISMA require written permission from the rights holder.

Author and initial rights holder: Roger Navarro. Brand/research line: Rogex Laboratories / Rogex PRISMA. The license does not grant trademark rights to Rogex Laboratories, Rogex PRISMA, PRISMA, logos, or related branding.

License summary

Free for research and learning. Not free to resell, repackage, embed in commercial products, or offer as a paid service without explicit written permission.

10. Scientific limitations

- PRISMA 3 is not a medical device.
- PRISMA 3 does not diagnose disease or provide clinical decisions.
- PRISMA 3 does not directly measure neurotransmitters, CB1, CB2 or the endocannabinoid system.
- ec/eo is a robust and relatively easy EEG contrast; it does not prove performance on difficult mental-state classification tasks.
- N=28 is useful as an initial validation, but stronger publication-grade claims require larger cohorts, confidence intervals, per-subject precision, test-retest evaluation and external review.
- The personalized intra-CV result is easier than LOSO and must not be presented as cross-subject generalization.
- This validation run does not include full artifact rejection/ICA.
- All outputs must be interpreted with signal quality, uncertainty, artifacts and methodological limits.

11. Roadmap

Phase	Target
PRISMA 3 public release	Documented research software, real EEG validation appendix, public web, README and research license.
PRISMA 4	Live EEG ingestion, improved artifact handling, larger real-dataset benchmarks and calibration workflow.
Research collaboration	External validation, protocol design, university/lab review and reproducibility packages.
Commercial path	Only after scientific review, IP clarity, licensing structure and regulatory boundary assessment.

Appendix A - Source package inventory

The PRISMA 3 source package contains Python modules for simulation, feature extraction, baseline modelling, individual translation, model evaluation, real EEG ingestion, documentation, tests, configuration and generated reports. The package should be cleaned before publication by excluding `__pycache__`, `.git` directories inside datasets, private data, and local cache folders.

Appendix B - Reference documents

- `real_eeg_validation_report.md` - supervised real EEG ec/eo appendix on OpenNeuro DS007358.
- `model_report.md` - synthetic five-state demo classification report.
- `HOW_IT_WORKS.md` - PRISMA 3 pipeline documentation.
- `HOW_TO_CONNECT_EEG.md` - CSV, MNE, EEGDash/OpenNeuro and LSL ingestion notes.
- `INTERINDIVIDUAL_VARIABILITY.md` - central methodological problem statement.
- `SCIENTIFIC_LIMITS.md` - non-medical, non-diagnostic limits.